

Mathematical Foundations of Reinforcement Learning and Online Decision Making

Lecture 6

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Overview

- 1 Synchronous TD Learning
- 2 Asynchronous TD Learning
- 3 Policy Iteration: Evaluation + Improvement
- 4 Q-learning: Value Iteration From Samples

What Is Synchronous TD Learning?

At iteration k : For every state $s \in \mathcal{S}$:

- 1 Generate one transition

$$S'_k(s) \sim P(\cdot | s).$$

- 2 Observe a reward $R_k(s)$.
- 3 Update the value of state s :

$$v_{k+1}(s) = (1 - \alpha)v_k(s) + \alpha \left(R_k(s) + \gamma v_k(S'_k(s)) \right).$$

One iteration = one sampled Bellman update per state

At each time step: for every state sample next state and reward.

The Sampled Bellman Operator

Define

$$\widehat{\mathcal{T}}_k v = \begin{bmatrix} R_k(1) + \gamma v(S'_k(1)) \\ \vdots \\ R_k(n) + \gamma v(S'_k(n)) \end{bmatrix}.$$

Since

$$\mathbb{E}[R_k(s)] = r(s) \quad \text{and} \quad \mathbb{E}[v(S'_k(s))] = \sum_{s'} P(s, s') v(s'),$$

we obtain

$$\mathbb{E}[\widehat{\mathcal{T}}_k v] = r + \gamma P v = \mathcal{T} v.$$

Define the noise $\xi_{k+1} = \widehat{\mathcal{T}}_k v_k - \mathcal{T} v_k$ then

$$\widehat{\mathcal{T}}_k v_k = \mathcal{T} v_k + \xi_{k+1}$$

with $\mathbb{E}[\xi_{k+1} \mid v_k] = 0$.

Variance of the Bellman Noise

For each state,

$$\xi_{k+1}(s) = R_k(s) - r(s) + \gamma \left(v_k(S'_k(s)) - \sum_{s'} P(s, s') v_k(s') \right).$$

Hence

$$\text{Var}(\xi_{k+1}(s) \mid v_k) = \text{Var}(R_k(s)) + \gamma^2 \text{Var}(v_k(S'_k(s))).$$

If rewards are bounded and

$$\|v_k\|_\infty \leq V_{\max},$$

then

$$\text{Var}(\xi_{k+1}(s) \mid v_k) \leq \sigma^2$$

for some constant σ^2 . Therefore

$$\boxed{\mathbb{E}[\|\xi_{k+1}\|^2 \mid v_k] \leq \sigma^2.}$$

Synchronous TD Learning for an MRP

Consider a finite Markov reward process with

$$\mathcal{T}v = r + \gamma P v, \quad v^* = \mathcal{T}v^*,$$

where $\gamma \in (0, 1)$ and P is a transition matrix. At iteration k , suppose that we obtain a noisy Bellman observation

$$\hat{\mathcal{T}}_k v_k = \mathcal{T}v_k + \xi_{k+1},$$

where

$$\mathbb{E}[\xi_{k+1} \mid \mathcal{F}_k] = 0.$$

The synchronous TD update is

$$v_{k+1} = (1 - \alpha)v_k + \alpha \hat{\mathcal{T}}_k v_k,$$

with a fixed step size $\alpha \in (0, 1]$.

Error Dynamics

Let

$$e_k = v_k - v^*.$$

Using $v^* = \mathcal{T}v^*$, we obtain

$$\begin{aligned} e_{k+1} &= (1 - \alpha)e_k + \alpha(\mathcal{T}v_k - \mathcal{T}v^*) + \alpha\xi_{k+1} \\ &= ((1 - \alpha)I + \alpha\gamma P)e_k + \alpha\xi_{k+1}. \end{aligned}$$

Define

$$A_\alpha = (1 - \alpha)I + \alpha\gamma P.$$

Since $\|Px\|_\infty \leq \|x\|_\infty$,

$$\|A_\alpha x\|_\infty \leq \rho_\alpha \|x\|_\infty, \quad \rho_\alpha = 1 - \alpha(1 - \gamma) < 1.$$

Thus fixed-step-size TD is a contractive linear system driven by noise.

Initial Error and Accumulated Noise

Repeated substitution gives

$$e_k = A_\alpha^k e_0 + \alpha \sum_{t=0}^{k-1} A_\alpha^{k-1-t} \xi_{t+1}.$$

The error therefore consists of two terms:

$$e_k = \underbrace{A_\alpha^k e_0}_{\text{initial error}} + \underbrace{\alpha \sum_{t=0}^{k-1} A_\alpha^{k-1-t} \xi_{t+1}}_{\text{accumulated noise}}.$$

The initial error decays geometrically:

$$\|A_\alpha^k e_0\|_\infty \leq \rho_\alpha^k \|e_0\|_\infty.$$

Finite-Time Mean-Square Error

We have $\|A_\alpha x\| \leq \rho_\alpha \|x\|$ and $\mathbb{E}[\|\xi_{k+1}\|^2 \mid \mathcal{F}_k] \leq \sigma^2$. Since the noise is conditionally mean zero,

$$\mathbb{E}\|e_{k+1}\|^2 \leq \rho_\alpha^2 \mathbb{E}\|e_k\|^2 + \alpha^2 \sigma^2.$$

Iterating this recursion gives

$$\begin{aligned}\mathbb{E}\|e_k\|^2 &\leq \rho_\alpha^{2k} \|e_0\|^2 + \alpha^2 \sigma^2 \sum_{t=0}^{k-1} \rho_\alpha^{2t} \\ &= \rho_\alpha^{2k} \|e_0\|^2 + \frac{\alpha^2 \sigma^2 (1 - \rho_\alpha^{2k})}{1 - \rho_\alpha^2}.\end{aligned}$$

Using $\rho_\alpha = 1 - \alpha(1 - \gamma)$,

$$\frac{\alpha^2}{1 - \rho_\alpha^2} = \frac{\alpha}{(1 - \gamma)(2 - \alpha(1 - \gamma))}.$$

Step Size Trade-Off and Time Complexity

The mean-square error satisfies

$$\mathbb{E}\|e_k\|^2 \leq \underbrace{\rho_\alpha^{2k}\|e_0\|^2}_{\text{transient error}} + \underbrace{\frac{\alpha\sigma^2}{(1-\gamma)(2-\alpha(1-\gamma))}}_{\text{noise floor}}.$$

For small α ,

$$\text{noise floor} \approx \frac{\alpha\sigma^2}{2(1-\gamma)} = O\left(\frac{\alpha\sigma^2}{1-\gamma}\right).$$

To make the transient term at most ε , i.e. $\rho_\alpha^{2k}\|e_0\|^2 \leq \varepsilon$, for small $\alpha(1-\gamma)$,

$$k \approx \frac{1}{2\alpha(1-\gamma)} \log\left(\frac{\|e_0\|^2}{\varepsilon}\right).$$

smaller $\alpha \implies$ smaller error floor but slower convergence.

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Asynchronous TD Learning

Suppose we observe a single trajectory

$$S_0, R_0, S_1, R_1, S_2, \dots$$

generated by the MRP. At iteration k , we observe

$$S_k, \quad R_k, \quad S_{k+1},$$

and update only the value of the visited state:

$$v_{k+1}(S_k) = v_k(S_k) + \alpha (R_k + \gamma v_k(S_{k+1}) - v_k(S_k)).$$

For every $s \neq S_k$,

$$v_{k+1}(s) = v_k(s).$$

One iteration = one transition and one state update

A Vector Formulation

Let u_s denote the unit vector corresponding to state s . Define the TD error

$$\delta_k = R_k + \gamma v_k(S_{k+1}) - v_k(S_k).$$

Then asynchronous TD can be written as

$$v_{k+1} = v_k + \alpha u_{S_k} \delta_k.$$

Equivalently,

$$v_{k+1} = v_k + \alpha u_{S_k} (R_k + \gamma v_k(S_{k+1}) - v_k(S_k)).$$

Only the coordinate corresponding to S_k changes.

The Expected TD Update

Conditioned on $S_k = s$ and v_k ,

$$\begin{aligned}\mathbb{E}[\delta_k \mid S_k = s, v_k] &= r(s) + \gamma \sum_{s'} P(s, s') v_k(s') - v_k(s) \\ &= (\mathcal{T} v_k)(s) - v_k(s).\end{aligned}$$

Therefore,

$$\mathbb{E}[u_{S_k} \delta_k \mid S_k = s, v_k] = u_s((\mathcal{T} v_k)(s) - v_k(s)).$$

Thus TD performs a sampled Bellman correction, but only at the visited state.

Average Dynamics Under Stationarity

Assume that the Markov chain has stationary distribution μ , and define

$$D_\mu = \text{diag}(\mu(1), \dots, \mu(n)).$$

If $S_k \sim \mu$, then

$$\begin{aligned}\mathbb{E}[u_{S_k} \delta_k \mid v_k] &= D_\mu(\mathcal{T}v_k - v_k) \\ &= D_\mu(r + \gamma P v_k - v_k).\end{aligned}$$

Hence the average update is

$$\boxed{\mathbb{E}[v_{k+1} - v_k \mid v_k] = \alpha D_\mu(\mathcal{T}v_k - v_k).}$$

The fixed point is unchanged:

$$D_\mu(\mathcal{T}v - v) = 0 \iff v = v^*,$$

provided $\mu(s) > 0$ for every state.

Error Dynamics

Let

$$e_k = v_k - v^*.$$

Using

$$v^* = r + \gamma P v^*,$$

the average error dynamics are

$$\mathbb{E}[e_{k+1} \mid e_k] = (I - \alpha D_\mu (I - \gamma P)) e_k.$$

Define

$$A_{\text{async}} = I - \alpha D_\mu (I - \gamma P).$$

Then

$$\boxed{\mathbb{E}[e_{k+1} \mid e_k] = A_{\text{async}} e_k.}$$

The update frequency of each state now enters through D_μ .

A Weighted Norm

Define

$$\|x\|_{D_\mu}^2 = x^\top D_\mu x.$$

The matrix governing the mean dynamics is

$$A_{\text{async}} = I - \alpha D_\mu (I - \gamma P).$$

Unlike the synchronous case, a contraction argument in $\|\cdot\|_\infty$ is not immediate. Instead, consider

$$\|e_{k+1}\|_{D_\mu}^2.$$

The matrix

$$D_\mu (I - \gamma P)$$

is positive definite, and therefore there exists $\lambda > 0$ such that

$$x^\top D_\mu (I - \gamma P) x \geq \lambda \|x\|_{D_\mu}^2.$$

A Positive Definite Drift

Let

$$M = D_\mu(I - \gamma P).$$

Since $\mu(s) > 0$ for all states,

$$D_\mu \succ 0.$$

Moreover,

$$x^\top Mx = x^\top D_\mu(I - \gamma P)x \geq \lambda \|x\|_{D_\mu}^2$$

for some constant $\lambda > 0$. Thus the Bellman error generates a drift towards v^* .

$$x^\top D_\mu(I - \gamma P)x \geq \lambda \|x\|_{D_\mu}^2$$

Convergence of the Mean Dynamics

Ignoring the sampling noise,

$$e_{k+1} = (I - \alpha M)e_k.$$

Consider the weighted norm

$$\|e\|_{D_\mu}^2 = e^\top D_\mu e.$$

For sufficiently small α ,

$$\|e_{k+1}\|_{D_\mu}^2 = \|(I - \alpha M)e_k\|_{D_\mu}^2 \leq (1 - \alpha\lambda)\|e_k\|_{D_\mu}^2.$$

Therefore

$$\|e_k\|_{D_\mu}^2 \leq (1 - \alpha\lambda)^k \|e_0\|_{D_\mu}^2.$$

Iteration Complexity

To achieve

$$\|e_k\|_{D_\mu}^2 \leq \varepsilon,$$

it suffices that

$$(1 - \alpha\lambda)^k \|e_0\|_{D_\mu}^2 \leq \varepsilon.$$

Equivalently,

$$k \geq \frac{\log\left(\|e_0\|_{D_\mu}^2 / \varepsilon\right)}{-\log(1 - \alpha\lambda)}.$$

For small step sizes,

$$k = O\left(\frac{1}{\alpha\lambda} \log \frac{1}{\varepsilon}\right).$$

Synchronous vs Asynchronous TD

Synchronous: $k = O\left(\frac{1}{\alpha(1-\gamma)} \log \frac{1}{\varepsilon}\right).$

Asynchronous: $k = O\left(\frac{1}{\alpha\lambda} \log \frac{1}{\varepsilon}\right).$

The constant λ depends on

- the stationary distribution μ ,
- the transition matrix P ,
- how quickly all states are revisited.

States that are rarely visited slow down learning.

The Difficulty with a Single Trajectory

For synchronous TD, fresh samples are available at every iteration.
For a single trajectory,

$$S_{k+1} \sim P(\cdot | S_k),$$

and the iterate

$$v_k$$

depends on the entire past trajectory

$$S_0, S_1, \dots, S_k.$$

Therefore,

$$S_k \quad \text{and} \quad v_k$$

are not independent. As a result,

$$\mathbb{E}[u_{S_k} \delta_k \mid v_k] \neq D_\mu(\mathcal{T} v_k - v_k).$$

The convergence analysis must control this bias.

Mixing Controls the Bias

Let μ denote the stationary distribution. After sufficiently many steps,

$$P^t(s, \cdot) \approx \mu(\cdot).$$

The mixing time τ_{mix} measures how quickly this occurs. If the chain mixes rapidly, then the distribution of the current state is close to μ :

$$\mathbb{P}(S_k \in \cdot) \approx \mu(\cdot).$$

Consequently,

$$\mathbb{E}[u_{S_k} \delta_k \mid v_k] \approx D_\mu(\mathcal{T} v_k - v_k).$$

Fast mixing $\implies$ small Markovian bias.

Convergence Rate with Markovian Sampling

Under

- bounded rewards,
- ergodicity,
- sufficiently fast mixing,
- a small constant step size α ,

TD satisfies

$$\mathbb{E} \|v_k - v^*\|_{D_\mu}^2 \leq C_1(1 - c\alpha)^k \|v_0 - v^*\|_{D_\mu}^2 + C_2\alpha.$$

Compared with the synchronous case:

- geometric decay remains,
- the error floor remains $O(\alpha)$,
- the constants now depend on the mixing time.

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From Prediction to Control

So far:

$$v^\pi(s) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s \right].$$

Prediction:

Given π , estimate v^π .

Control:

Find an optimal policy $\pi^\star$.

Key challenge:

$\pi^\star$ is unknown.

We therefore alternate between

- evaluating the current policy,
- improving the policy using the value estimates.

Two Fundamental Operations

Policy Evaluation Estimate

$$v^\pi$$

for the current policy. **Policy Improvement** Construct a new policy

$$\pi'$$

that is greedy with respect to the current value function:

$$\pi'(s) = \arg \max_a q^\pi(s, a).$$

Policy Improvement Theorem:

$$q^\pi(s, \pi'(s)) \geq v^\pi(s)$$

for all states. Therefore,

$$v^{\pi'}(s) \geq v^\pi(s).$$

Generalized Policy Iteration (GPI)

Two processes interact:

Policy Evaluation

improves the value estimates

$$V \approx v^\pi.$$

Policy Improvement

improves the policy

$$\pi.$$

$$\pi \longrightarrow V^\pi \longrightarrow \text{Greedy}(\pi) \longrightarrow V \longrightarrow \dots$$

Neither process needs to finish before the other begins.

Policy Iteration as GPI

Policy Iteration alternates between

Evaluate π_k

and

Improve π_k .

$v^{\pi_k} \leftarrow \text{Policy Evaluation}$

$\pi_{k+1} \leftarrow \text{Greedy}(v^{\pi_k})$.

Each improvement step satisfies

$$v^{\pi_{k+1}} \geq v^{\pi_k}.$$

For finite MDPs,

$$\pi_k \rightarrow \pi^*.$$

GPI is the Foundation of RL Control

Dynamic Programming:

exact evaluation + greedy improvement.

Monte Carlo Control:

sample-based evaluation + greedy improvement.

TD Control:

bootstrapped evaluation + greedy improvement.

Different algorithms differ in how they perform

Policy Evaluation.

The policy improvement step is essentially the same.

Control = Evaluation + Improvement

Approximate Policy Improvement

Exact policy evaluation:

$$\pi' = \text{Greedy}(q^\pi)$$

implies

$$v^{\pi'} \geq v^\pi.$$

Approximate policy evaluation:

$$\|\hat{q} - q^\pi\|_\infty \leq \varepsilon$$

does not guarantee monotonic improvement. A small estimation error can change the greedy action. However, if ε is small, the resulting policy is approximately improving.

Approximate evaluation + greedy improvement $\approx$ policy improvement.

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From Prediction to Control

For policy evaluation, the policy is fixed:

$$Q^\pi = \mathcal{T}^\pi Q^\pi.$$

For control, we want to find

$$Q^*(s, a) = \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t R_t \mid S_0 = s, A_0 = a \right].$$

Once Q^* is known, an optimal policy is obtained by

$$\pi^*(s) \in \arg \max_a Q^*(s, a).$$

The key idea of Q-learning is to learn Q^* directly, without first learning the value of a fixed policy.

The Optimal Bellman Operator

Define the optimal Bellman operator

$$(\mathcal{T}^*Q)(s, a) = r(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q(s', a').$$

The optimal action-value function satisfies

$$Q^* = \mathcal{T}^*Q^*.$$

Thus Q^* is a fixed point of the optimal Bellman operator. Once Q^* is known, an optimal policy is

$$\pi^*(s) \in \arg \max_a Q^*(s, a).$$

The Optimal Bellman Operator Is a Contraction

For any two action-value functions Q_1 and Q_2 ,

$$\begin{aligned} & |(\mathcal{T}^* Q_1)(s, a) - (\mathcal{T}^* Q_2)(s, a)| \\ & \leq \gamma \sum_{s'} P(s'|s, a) \left| \max_{a'} Q_1(s', a') - \max_{a'} Q_2(s', a') \right|. \end{aligned}$$

Using

$$\left| \max_a x_a - \max_a y_a \right| \leq \max_a |x_a - y_a|,$$

we obtain

$$\|\mathcal{T}^* Q_1 - \mathcal{T}^* Q_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty.$$

Therefore $\mathcal{T}^*$ has a unique fixed point Q^* .

What Is Synchronous Q-Learning?

At iteration k , for every $(s, a) \in \mathcal{S} \times \mathcal{A}$:

- 1 Generate one transition

$$S'_k(s, a) \sim P(\cdot | s, a).$$

- 2 Observe a reward $R_k(s, a)$.

- 3 Update

$$Q_{k+1}(s, a) = (1-\alpha)Q_k(s, a) + \alpha \left(R_k(s, a) + \gamma \max_{a'} Q_k(S'_k(s, a), a') \right)$$

One iteration = one sampled Bellman update per state-action pair.

A Sampled Optimal Bellman Operator

Define

$$(\hat{\mathcal{T}}_k^* Q)(s, a) = R_k(s, a) + \gamma \max_{a'} Q(S'_k(s, a), a').$$

Then synchronous Q-learning becomes

$$Q_{k+1} = (1 - \alpha)Q_k + \alpha \hat{\mathcal{T}}_k^* Q_k.$$

Furthermore,

$$\begin{aligned} \mathbb{E}[(\hat{\mathcal{T}}_k^* Q)(s, a)] &= r(s, a) + \gamma \sum_{s'} P(s'|s, a) \max_{a'} Q(s', a') \\ &= (\mathcal{T}^* Q)(s, a). \end{aligned}$$

Thus

$$\mathbb{E}[\hat{\mathcal{T}}_k^* Q] = \mathcal{T}^* Q.$$

Optimal Bellman Operator Plus Noise

Define the Bellman noise

$$\xi_{k+1} = \hat{\mathcal{T}}_k^* Q_k - \mathcal{T}^* Q_k.$$

Then

$$\boxed{\hat{\mathcal{T}}_k^* Q_k = \mathcal{T}^* Q_k + \xi_{k+1}.$$

Conditioned on Q_k ,

$$\mathbb{E}[\xi_{k+1} \mid Q_k] = 0.$$

Therefore, Q-learning can be written as

$$Q_{k+1} = (1 - \alpha)Q_k + \alpha(\mathcal{T}^* Q_k + \xi_{k+1}).$$

It is a stochastic approximation of the fixed-point iteration

$$Q \leftarrow \mathcal{T}^* Q.$$

What Is the Bellman Noise?

For each state-action pair,

$$\xi_{k+1}(s, a) = R_k(s, a) - r(s, a) + \gamma \left(\max_{a'} Q_k(S'_k(s, a), a') - \sum_{s'} P(s'|s, a) \max_{a'} Q_k(s', a') \right).$$

The noise comes from two sources:

- randomness in the observed reward,
- randomness in the sampled next state.

Under bounded rewards and bounded iterates, there exists $\sigma^2 < \infty$ such that

$$\mathbb{E}[\|\xi_{k+1}\|_\infty^2 \mid Q_k] \leq \sigma^2.$$

Boundedness of the Q-Learning Iterates

Assume that

$$|R_k(s, a)| \leq R_{\max}$$

and define

$$B = \max \left\{ \|Q_0\|_{\infty}, \frac{R_{\max}}{1 - \gamma} \right\}.$$

Suppose that $\|Q_k\|_{\infty} \leq B$. Then

$$\begin{aligned} \|Q_{k+1}\|_{\infty} &\leq (1 - \alpha)\|Q_k\|_{\infty} + \alpha(R_{\max} + \gamma\|Q_k\|_{\infty}) \\ &\leq (1 - \alpha)B + \alpha(R_{\max} + \gamma B) \\ &\leq B. \end{aligned}$$

Therefore, by induction,

$$\|Q_k\|_{\infty} \leq B \quad \text{for every } k.$$

Hence the Bellman noise has uniformly bounded variance.

The Q-Learning Error Recursion

Let

$$e_k = Q_k - Q^*.$$

Using

$$Q^* = \mathcal{T}^* Q^*,$$

we obtain

$$\begin{aligned} e_{k+1} &= (1 - \alpha)e_k \\ &\quad + \alpha(\mathcal{T}^* Q_k - \mathcal{T}^* Q^*) + \alpha\xi_{k+1}. \end{aligned}$$

Since $\mathcal{T}^*$ is a contraction,

$$\|\mathcal{T}^* Q_k - \mathcal{T}^* Q^*\|_\infty \leq \gamma \|e_k\|_\infty.$$

Therefore,

$$\|e_{k+1}\|_\infty \leq \rho_\alpha \|e_k\|_\infty + \alpha \|\xi_{k+1}\|_\infty,$$

where $\rho_\alpha = 1 - \alpha(1 - \gamma)$.

Convergence Without Sampling Noise

First, suppose that $\xi_{k+1} = 0$. Then

$$Q_{k+1} = (1 - \alpha)Q_k + \alpha \mathcal{T}^* Q_k.$$

The error satisfies

$$\|e_{k+1}\|_\infty \leq \rho_\alpha \|e_k\|_\infty,$$

where

$$\rho_\alpha = 1 - \alpha(1 - \gamma).$$

Iterating gives

$$\|Q_k - Q^*\|_\infty \leq \rho_\alpha^k \|Q_0 - Q^*\|_\infty.$$

Thus the deterministic iteration converges geometrically to Q^* .

Deterministic Iteration Complexity

To guarantee

$$\|Q_k - Q^*\|_\infty \leq \varepsilon,$$

it is sufficient that

$$\rho_\alpha^k \|Q_0 - Q^*\|_\infty \leq \varepsilon.$$

Therefore,

$$k \geq \frac{\log(\|Q_0 - Q^*\|_\infty / \varepsilon)}{-\log \rho_\alpha}.$$

Since

$$-\log \rho_\alpha \approx \alpha(1 - \gamma),$$

we obtain

$$k = O\left(\frac{1}{\alpha(1 - \gamma)} \log \frac{1}{\varepsilon}\right).$$

Fixed Step Size and Sampling Noise

With sampling noise,

$$Q_{k+1} = (1 - \alpha)Q_k + \alpha(\mathcal{T}^*Q_k + \xi_{k+1}).$$

The contraction repeatedly reduces previous errors, while every iteration introduces new noise of size $\alpha\xi_{k+1}$. Consequently, the error has two components:

error at time k = transient error + accumulated sampling noise.

The transient error decreases geometrically at rate

$$\rho_\alpha = 1 - \alpha(1 - \gamma).$$

The accumulated noise does not vanish when α is fixed.

Finite-Time Error Structure

Under bounded rewards and independent synchronous samples, fixed-step-size Q-learning admits a bound of the form

$$\mathbb{E} [\|Q_k - Q^*\|_\infty^2] \leq C_1 \rho_\alpha^{2k} \|Q_0 - Q^*\|_\infty^2 + C_2 \frac{\alpha \sigma^2}{1 - \gamma}.$$

Here

$$\rho_\alpha = 1 - \alpha(1 - \gamma),$$

and C_1, C_2 may depend on the number of state-action pairs and on the precise noise assumptions. Thus

$$\text{mean-square error} = \text{geometric transient} + O\left(\frac{\alpha \sigma^2}{1 - \gamma}\right).$$

The Fixed-Step-Size Error Floor

As $k \rightarrow \infty$, the transient term disappears:

$$\rho_{\alpha}^{2k} \|Q_0 - Q^*\|_{\infty}^2 \rightarrow 0.$$

However, the sampling noise remains:

$$\limsup_{k \rightarrow \infty} \mathbb{E} [\|Q_k - Q^*\|_{\infty}^2] = O\left(\frac{\alpha \sigma^2}{1 - \gamma}\right).$$

Hence fixed-step-size Q-learning does not generally converge exactly to Q^* . Instead, it fluctuates around Q^* .

smaller α $\implies$ smaller error floor.
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Time to Reach the Error Floor

To make the transient error smaller than ε , we require

$$\rho_\alpha^{2k} \|Q_0 - Q^*\|_\infty^2 \leq \varepsilon.$$

This gives

$$k \geq \frac{\log (\|Q_0 - Q^*\|_\infty^2 / \varepsilon)}{-2 \log \rho_\alpha}.$$

For small $\alpha(1 - \gamma)$,

$$k = O\left(\frac{1}{\alpha(1 - \gamma)} \log \frac{1}{\varepsilon}\right).$$

The noise variance does not change the contraction rate, but it determines how small an error can be reached.

Choosing the Step Size for a Target Accuracy

Suppose that we want

$$\mathbb{E} [\|Q_k - Q^*\|_\infty^2] \leq \varepsilon.$$

The error floor must first satisfy

$$\frac{\alpha \sigma^2}{1 - \gamma} \lesssim \varepsilon.$$

Thus we choose approximately

$$\alpha = O\left(\frac{\varepsilon(1 - \gamma)}{\sigma^2}\right).$$

Substituting this into the transient convergence time gives

$$k = O\left(\frac{\sigma^2}{\varepsilon(1 - \gamma)^2} \log \frac{1}{\varepsilon}\right).$$

Thus larger noise requires a smaller step size and more iterations.

What Determines the Convergence Rate?

The main factors are:

- **Step size α :**

larger $\alpha \Rightarrow$ faster contraction but larger error floor.

- **Discount factor γ :**

$\gamma \rightarrow 1 \Rightarrow$ slower contraction and larger error floor.

- **Noise variance σ^2 :**

larger $\sigma^2 \Rightarrow$ larger error floor.

- **Number of state-action pairs:** Controlling the maximum error over all (s, a) generally introduces dimension-dependent constants.

$$\text{effective horizon} = \frac{1}{1 - \gamma}.$$

Synchronous Q-Learning: Main Conclusion

The optimal Bellman operator satisfies

$$\|\mathcal{T}^*Q_1 - \mathcal{T}^*Q_2\|_\infty \leq \gamma \|Q_1 - Q_2\|_\infty.$$

Therefore, synchronous Q-learning is a noisy contractive iteration:

$$Q_{k+1} = (1 - \alpha)Q_k + \alpha (\mathcal{T}^*Q_k + \xi_{k+1}).$$

With a fixed step size,

error = geometrically decaying transient + non-vanishing error floor.

Approximately,

$$\text{convergence time} \sim \frac{1}{\alpha(1 - \gamma)}, \quad \text{error floor} \sim \frac{\alpha\sigma^2}{1 - \gamma}.$$

Asynchronous Q-Learning

Suppose that we observe a single trajectory

$$S_0, A_0, R_0, S_1, A_1, R_1, \dots$$

At iteration k , only the visited state-action pair is updated:

$$Q_{k+1}(S_k, A_k) = Q_k(S_k, A_k) + \alpha \left(R_k + \gamma \max_a Q_k(S_{k+1}, a) - Q_k(S_k, A_k) \right).$$

For every $(s, a) \neq (S_k, A_k)$,

$$Q_{k+1}(s, a) = Q_k(s, a).$$

One transition = one state-action update.

A Vector Formulation

Let $u_{s,a}$ denote the unit vector corresponding to (s, a) . Define the Q-learning TD error

$$\delta_k = R_k + \gamma \max_a Q_k(S_{k+1}, a) - Q_k(S_k, A_k).$$

Then

$$Q_{k+1} = Q_k + \alpha u_{S_k, A_k} \delta_k.$$

Conditioned on $S_k = s$, $A_k = a$, and Q_k ,

$$\mathbb{E}[\delta_k \mid S_k = s, A_k = a, Q_k] = (\mathcal{T}^* Q_k)(s, a) - Q_k(s, a).$$

Thus Q-learning performs a sampled optimal Bellman correction at the visited state-action pair.

A Fixed Exploration Policy

Suppose actions are selected by a fixed exploration policy

$$\beta(a|s).$$

Then

$$(S_k, A_k)$$

forms a Markov chain on the state-action space. Assume that this chain is ergodic, with stationary distribution

$$d_\beta(s, a) > 0.$$

Define

$$D_\beta = \text{diag}(d_\beta(s, a)).$$

In stationarity, the average update has the form

$$\mathbb{E}[Q_{k+1} - Q_k] = \alpha D_\beta (\mathcal{T}^* Q_k - Q_k).$$

The Role of the Stationary Distribution

The factor

$$d_{\beta}(s, a)$$

is the long-run fraction of time that (s, a) is updated. State-action pairs with large $d_{\beta}(s, a)$ are updated frequently. State-action pairs with small $d_{\beta}(s, a)$ are updated rarely. Define

$$d_{\min} = \min_{s,a} d_{\beta}(s, a).$$

If $d_{\min}$ is small, some components of Q learn very slowly.

Good exploration $\implies$ all state-action pairs are updated sufficiently often

Why Is the Analysis More Difficult?

The iterate Q_k depends on the past trajectory

$$S_0, A_0, \dots, S_k, A_k.$$

Therefore,

$$(S_k, A_k) \quad \text{and} \quad Q_k$$

are generally dependent. Consequently,

$$\mathbb{E}[u_{S_k, A_k} \delta_k \mid Q_k] \neq D_\beta (\mathcal{T}^* Q_k - Q_k)$$

in general. The difference is a Markovian sampling bias.

The stationary average is not the exact conditional update.

Mixing Controls the Markovian Bias

Let d_β be the stationary distribution of the state-action Markov chain. The mixing time measures how quickly

$$\mathbb{P}((S_{k+t}, A_{k+t}) \in \cdot \mid S_k = s, A_k = a)$$

approaches d_β . If the exploration chain mixes rapidly, then after a moderate number of steps,

$$(S_{k+t}, A_{k+t}) \approx d_\beta.$$

Hence the actual trajectory update is close to the stationary average update.

Fast mixing $\implies$ smaller temporal-dependence effects.

Asynchronous Q-Learning: Convergence Picture

Under

- bounded rewards,
- sufficient exploration,
- ergodicity of the exploration chain,
- sufficiently fast mixing,
- a sufficiently small fixed step size,

the error has the same general structure:

$$\mathbb{E} [\|Q_k - Q^*\|^2] \leq \underbrace{C_1 \exp(-c\alpha d_{\min}(1-\gamma)k)}_{\text{transient}} + \underbrace{C_2 \alpha \sigma_{\text{eff}}^2}_{\text{error floor}}.$$

The constants depend on the visitation distribution and mixing time.

Synchronous and Asynchronous Q-Learning

Synchronous Q-learning

- every state-action pair is updated,
- fresh samples can be independent,
- analysis follows directly from Bellman contraction.

Asynchronous Q-learning

- only the visited state-action pair is updated,
- update frequencies depend on $d_\beta(s, a)$,
- consecutive samples are dependent,
- mixing controls the Markovian sampling effect.

Contraction + coverage + mixing

are the three main ingredients in the convergence analysis.